GLYCERYL TRINITRATE

Never with sildenafil/tadalafil

Check BP before dose

CHEMICAL NAME Propane-1,2,3-triyl trinitrate (nitroglycerin)
CLASS Organic nitrate vasodilator (antianginal)
TRADE NAMES Nitrolingual (spray) · GTN tablets · Nitro-Dur / Transiderm-Nitro (patch) · Tridil (IV)

THERAPEUTIC USES / INDICATIONS

- Acute angina attack (sublingual).
- Prophylaxis before exertion.
- Acute coronary syndrome, acute heart failure / pulmonary oedema (IV).
- Hypertensive emergency (perioperative); anal fissure (ointment).

ADMINISTRATION

- **SL tablet/spray:** sit down; place under tongue / spray onto or under tongue — do not inhale spray.
- **IV infusion:** via syringe pump using non-PVC (polyethylene) giving set (GTN absorbs into PVC).
- **Patch:** apply daily, remove for 8–12 h nitrate-free interval (prevents tolerance).
- Tablets: store in original glass bottle; discard 8 weeks after opening.

DOSAGES

- **Acute angina:** 0.3–0.6 mg SL tablet or 1–2 sprays (400 mcg each); repeat every 5 min, max 3 doses in 15 min — call emergency services if pain persists.
- **IV:** 10–200 mcg/min (start 5–10 mcg/min, titrate every 3–5 min).
- **Patch:** start 5 mg/24 h, titrate as needed; patch-free interval of 8–12 h (usually at night).
- **Ointment (anal fissure):** 0.4% twice daily.

PHARMACOKINETICS

- **Onset:** SL 1–3 min; IV immediate. **Duration:** SL 20–30 min; patch while applied.
- Extensive first-pass metabolism (not effective swallowed).
- Metabolised in liver and blood vessels. **t½:** 1–4 min.

NURSING MANAGEMENT

- Check BP before each dose; withhold if SBP <90.
- Patient sits or lies down before taking.
- Record pain score before and after each dose; 12-lead ECG during chest pain.
- IV: BP every 5–15 min during titration; use non-PVC set.
- Remove patch at night (nitrate-free period); rotate sites; remove before defibrillation/MRI.

PHARMACODYNAMICS

- Releases **nitric oxide** → ↑ cGMP → smooth muscle relaxation.
- Venodilation (↓ preload) > arterial dilation; coronary vasodilation and relief of spasm.
- ↓ myocardial oxygen demand.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Headache (throbbing), flushing.
- Dizziness, postural hypotension.
- Reflex tachycardia.

ADVERSE EFFECTS (SERIOUS)

- Severe hypotension, syncope.
- Tolerance with continuous use.
- Methaemoglobinaemia (high doses).
- Rebound angina on abrupt withdrawal of long-term therapy.

CONTRAINDICATIONS & CAUTIONS

- Use of PDE-5 inhibitors (sildenafil, vardenafil within 24 h; tadalafil within 48 h) — fatal hypotension.
- SBP <90 mmHg; hypovolaemia.
- Right ventricular infarction; severe aortic stenosis, hypertrophic obstructive cardiomyopathy.
- Raised intracranial pressure, constrictive pericarditis, cardiac tamponade.

KEY INTERACTIONS

- **Sildenafil, tadalafil, vardenafil, riociguat:** severe hypotension; contraindicated.
- **Alcohol, antihypertensives:** ↑ hypotension.
- **Heparin:** IV GTN may ↓ heparin effect.

PATIENT EDUCATION

- Sit down before using; a burning/tingling and headache mean it is working.
- If pain persists 5 min after the first dose, call an ambulance (and take a second dose).
- Never use with erectile dysfunction tablets.
- Carry it always; keep tablets in their original bottle.

KEY POINTS

- Nitrate; NO donor; venodilator.
- Never with sildenafil/tadalafil.
- SL onset 1–3 min; max 3 doses.
- Check BP; headache common.

SOURCES CHECKED [353] Glyceryl Trinitrate 1mg/ml solution for infusion (UK SmPC) [354] Nitroglycerin, IV (US label) [355] Nitrostat, SL (US label) *Full references p. 432–457*